Artificial Intelligence Transformation: Proposal for a Fintech Startup

1. Introduction

A wave of digitalisation and open banking regulations, including the Payment Services Directive 2 (PSD2), has unleashed fintech as a transformative force in finance, unlocking widespread access to transaction data and fueling analytics-driven innovation (European Union, 2015; McKinsey & Company, 2023).

Up to 90% of fintechs globally leverage artificial intelligence (AI), harnessing high-frequency data to automate decisions, gain insights, and cut costs—reshaping workflows and competition (Dimension Market Research, 2024; McKinsey & Company, 2025).

Over 70% of AI-investing fintechs report greater profitability, higher productivity, better customer satisfaction, and lower costs (World Economic Forum, 2025). Today, AI is essential for growth, innovation, competitiveness, and customer trust.

This report introduces a B2B analytics startup and proposes three machine learning (ML) business cases to enhance its rules-based (RB) mechanics: transaction categorisation, recurring expense identification, and financial health scoring. It outlines data needs, AI solutions using Random Forest (RF) and Gradient Boosting (GB), a testing methodology, and key technological, ethical, legal, and social considerations to inform AI adoption.

2. Startup Context

AnalyseWisely, a four-year-old hypothetical B2B fintech startup, is locally established in the Dutch market and provides an intelligence layer and analytical insights. It uses

a licensed third-party open banking provider, like SaltEdge (n.d.), for secure, compliant API access to customer bank data under PSD2.

The startup excels at transforming raw financial data into clean, structured, analytics-ready datasets. Its core value is providing categorisation and data-enrichment insights, delivering foundational intelligence to improve client efficiency and enable data-driven decisions. It is also exploring early financial behavioural signals to expand its offerings. Currently, AnalyseWisely serves ten small- and medium-sized enterprises, including accounting firms, tax advisors, and fintech applications.

3. Business Complications and Opportunities

AnalyseWisely excels in data ingestion, translation, and cleaning, relying on a simple, interpretable If-Then rule for classification. While effective, cost-efficient, and regulatory-friendly, this approach is only moderately precise and requires ongoing expert maintenance (Arner, Barberis and Buckley, 2016; Jørgensen and Igel, 2021). Growing customer complexity, multilingual merchants, and higher security demands challenge these foundations (Capgemini, 2026). Reluctance to adapt risks scalability, service, and growth. Falling behind in fintech AI trends weakens the value proposition, reduces retention, and increases the risk of becoming a commodity provider, risking market exit.

Fortunately, adopting AI does not require overhauling existing infrastructure. The current rule-based (RB) system, reliant on human expertise, can be augmented with AI to improve pattern recognition and predictive accuracy. Embracing new technology can help AnalyseWisely stand out as a forward-thinking, trustworthy company that combines automation and human judgment. This will also support a smooth future

transition from open banking to open finance (OECD, 2023), unlocking broader access to financial data and new monetisation opportunities.

4. AI Solutions Proposal

This proposal aims to enhance RB analytics by applying supervised ML to three solutions: transactional categorisation, recurring expense identification, and financial health scoring, each adding proven fintech analytical sophistication.

a. Transactional Categorisation

Transactional categorisation assigns transactions to labels such as rent, transport, subscriptions, utilities, and travel, based on context, purpose, and features like merchant details, dates, and amounts. This enables efficient classification, supports adaptive pattern learning, and reduces compliance risks by preventing misinformation in financial reports and audits (García-Méndez *et al*., 2020; Jørgensen and Igel, 2021; Kotios *et al*., 2022; Lesner *et al.*, 2019; Ta, Ben Saad and Oh, 2023).

b. Recurring Payment Detection

Recurring payment detection identifies frequent expenses by finding repeated transactions with consistent timing, amounts, and merchants. This increases visibility of recurring costs, predicts future obligations, supports budgeting, and flags unnecessary subscriptions to prevent losses (Alamri and Ykhlef, 2024; Fulcher and Jones, 2014; Lee, Lee and Kim, 2023; Li, Ren and Wu, 2025).

c. Behavioural Financial Health Scoring

Financial health scoring uses factors like income stability, spending patterns, and recurring obligations to assess customers' financial health. This enables actionable

budgeting insights, early risk detection with proactive, personalised intervention, and efficient segmentation with strong return on investment (Agrawal, Khanna and Hamdare, 2025; Butler *et al*., 2023; Kotios *et al*., 2022; Lee *et al*., 2023; Pragasam *et al*., 2023; Whitaker, 2025).

5. Data Requirements

With secure PSD2 access to transaction-level data via a third-party provider, AnalyseWisely can already access customer records. Given the data's sensitivity, privacy-by-design measures must be applied, ensuring that personally identifiable data is masked and tokenised to prevent exposure and to guarantee individuals' protection before processing and analysis (European Union, 2016). Raw data fields such as description, amount, date, location, merchant, and category are sufficient to test the three proposed ML use cases.

6. AI Implementation Approach

AI Implementation will be tested via the AI Pipeline using the CRISP-DM framework (Data Science Process Alliance, n.d.), which offers structured technical guidance aligned with business requirements for data mining projects (Schröer, Kruse, and Gómez, 2021). The process starts with understanding the business—defining objectives, scope, goals, KPIs, and success metrics for each use case to ensure stakeholder alignment.

After obtaining and ensuring privacy of transactional data, the next steps are data preparation (cleaning, imputing, formatting), followed by feature engineering. This phase—generating, transforming, and selecting features—directly impacts data mining success (Doube, 2022). Guidance and financial expertise ensure key features

are well-defined and validated during model training. In the modelling stage, suitable machine learning techniques are applied to each task.

Finally, insights from model training and testing are evaluated for alignment with predefined business goals. This informs whether to proceed with live deployment or refine the approach by revisiting earlier pipeline phases, as encouraged by the iterative, feedback-driven, and continuous-improvement nature of CRISP-DM.

7. Model Selection

The proposed ML approach uses two supervised ensemble methods—RF and GB—both of which require labelled data. These algorithms are ideal for structured tabular data with high dimensionality and often imbalanced distributions, and deliver strong performance on complex classification and regression tasks.

a. Random Forest

RF generates decision trees in parallel, randomly selecting features at each split to optimise accuracy and stability. Majority voting across these uncorrelated trees reduces overfitting, enhances robustness, and improves generalisation. RF handles complex datasets well, though large models may limit transparency. Feature importance analysis, like SHAP, aids interpretability. Valued for simplicity, rapid training, and minimal preprocessing, RF's strong performance and feature analysis have made it a standard in fields from credit and fraud detection to medical forecasting (Grinsztajn, Oyallon and Varoquaux, 2022; Mienye and Jere, 2024; Mienye and Sun, 2022; Pragasam *et al*., 2023; Russell and Norvig, 2021; Shwartz-Ziv and Armon, 2022).

b. Gradient Boosting

GB builds trees sequentially, with each iteration refining previous results to minimise errors and the loss function. While prone to overfitting due to its greedy gradient descent optimisation and higher computational cost, GB delivers robust, accurate predictions. Though often seen as opaque, post hoc feature-importance analysis enhances its interpretability. GB's strengths drive its popularity in fraud detection, stock prediction, credit scoring, medical diagnosis, and marketing applications like churn prediction and recommender systems (Dube and Verster, 2023; IBM, n.d.; Mienye and Jere, 2024; Mienye and Sun, 2022; Pragasam *et al.*, 2023; Russell and Norvig, 2021).

8. Potential Limitations and Risks of the AI Solution

The effectiveness of ML models relies heavily on open banking data, which is frequently non-standardised, inconsistent, and often lacks event context or critical values (García-Méndez *et al*., 2020; Shwartz-Ziv and Armon, 2022; Ta, Ben Saad, and Oh, 2023). Additionally, under PSD2 regulations, access to historical data is limited to 180 days (European Banking Authority, 2022). These challenges undermine the accuracy and reliability of predictions.

Imbalanced, high-dimensional data raise the risk of biased or discriminatory learning, especially with complex, overfitting-prone models. Such models may overemphasise dominant patterns and neglect minority behaviours, reducing generalisability (Altalhan, Algarni and Turki-Hadj Alouane, 2025; Lyu, Zhou and Zhong, 2025). In rapidly evolving financial environments, shifts in preferences, seasonality, merchant base, and data quality can trigger data drift, making models unable to adapt and leading to unreliable predictions (Pugliese, Costa, and Hirata, 2021).

Data quality outweighs model complexity (Shwartz-Ziv and Armon, 2022). Model performance depends on the quality and representativeness of the underlying dataset (Mohammed *et al.*, 2025). Algorithmic accuracy, reliability, and transparency depend on meticulous data provenance, rigorous data protection, preparation, and well-defined feature representation—the foundation of Trustworthy AI (Duboue, 2020; Grinsztajn, Oyallon, and Varoquaux, 2022; Theobald, 2024).

9. Challenges and Implications in Employing AI Technology

The sensitive nature of open banking financial data presents social, ethical, and legal challenges. PSD2 limits data access based on explicit user consent (European Union, 2015), and the GDPR requires data minimisation and anonymisation (European Union, 2016), resulting in a partial financial view. These necessary constraints protect individuals but may reduce model performance and obscure nuanced patterns, raising concerns about model fairness. When models are built on incomplete or unrepresentative data, algorithmic design can perpetuate and amplify bias—even in the most advanced and high-quality systems (Acharya, Divya, and Kuppan, 2024).

As a consequence, complex models can yield opaque outcomes with significant social impact. Transparency, explainability and safety are essential for responsible and trustworthy AI, as required by the GDPR and EU AI Act (European Union, 2024). Guided by the Code of Conduct, human-in-the-loop oversight enables proactive AI governance by helping financial experts detect and correct bias at all stages of AI development, support trade-off decisions, and ensure outcomes align with human values and uphold social responsibility.

10. Conclusion

In today's fast-paced fintech ecosystem, where risk appetite is high and competition is intense, embracing AI is a strategic move for innovation and growth. This report proposes an ML approach to analyse transactional data and predict spending habits, recurring obligations, and financial stability. Ensemble models—RF and GB—demonstrate the potential of high-performing algorithms, excelling with structured, high-dimensional financial data while maintaining interpretability through post hoc feature analysis. Leveraging ML as a decision-support tool to enhance AnalyseWisely's RB systems can yield operational improvements and empower financial experts to drive economic value—while steadfastly prioritising customer-centricity and building trust.

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